

Technical Project Overview

LeGrad-Inspired Temporal Explanations for Speech Emotion Recognition

gradient_based_speech_xai

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1 Project Overview

This project compares two approaches for explaining a pretrained speech emotion recognition classifier. SpeechXAI provides human-readable explanations based on word-aligned audio segments. The proposed LeGrad-inspired method will derive temporal relevance from gradients associated with the attention mechanisms of a wav2vec 2.0 encoder. The classification task uses speech samples and emotion annotations from IEMOCAP.

The quantitative evaluation uses deletion by silence masking. Important audio regions are replaced by silence while the original signal length is preserved, and the resulting confidence drop for the classifier’s original predicted class is measured. The central methodological decision is to compare word-level SpeechXAI explanations with time-bin or token-level LeGrad explanations under the same total occluded duration.

2 Sources and External References

- **Dataset: IEMOCAP.** The local data follows the five-session IEMOCAP release from the USC Signal Analysis and Interpretation Laboratory. The corpus contains speech, transcripts, and categorical emotion annotations. Project source: [USC IEMOCAP website](#). Dataset paper: [Busso et al., “IEMOCAP: Interactive Emotional Dyadic Motion Capture”](#).
- **Pretrained classifier.** The project uses [superb/wav2vec2-base-superb-er](#), the Hugging Face wav2vec 2.0 emotion-recognition checkpoint loaded by the SpeechXAI repository for IEMOCAP. The model exposes four output classes: anger, happiness/excitement, neutral, and sadness.
- **wav2vec 2.0.** The encoder architecture originates from Baevski et al., [“wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations”](#).
- **SUPERB.** The active checkpoint follows the SUPERB speech-processing benchmark described by Yang et al., [“SUPERB: Speech Processing Universal PERFORMANCE Benchmark”](#).
- **SpeechXAI.** The word-level explanation method follows Pastor et al., [“Explaining Speech Classification Models via Word-Level Audio Segments and Paralinguistic Features,” EACL 2024](#). Repository: [elianap/SpeechXAI](#).
- **LeGrad.** The proposed temporal method is inspired by Bousselham et al., [“LeGrad: An Explainability Method for Vision Transformers via Feature Formation Sensitivity”](#). Repository: [WalBouss/LeGrad](#).

The current project configuration uses IEMOCAP. No separate YAMLCap dataset source is configured.

3 Notebook-by-Notebook Technical Overview

3.1 Notebook 01 — SpeechXAI Word Attribution on One IEMOCAP Audio

Purpose

Establish a complete SpeechXAI explanation and deletion-evaluation workflow on one IEMOCAP utterance before integrating the LeGrad-inspired method.

Technical idea

SpeechXAI uses WhisperX to transcribe the utterance and align each word with its actual start and end time. The same timestamp objects are passed into SpeechXAI’s leave-one-out explainer, ensuring that each attribution score refers to the word interval shown in the visualization. SpeechXAI measures importance as the confidence change caused by replacing a word region with silence.

Decisions made

- A pretrained wav2vec 2.0 emotion classifier is used so the project can focus on explanation quality rather than classifier training.
- One utterance is used to keep this notebook focused on the explanation workflow and visualization.
- The original predicted class is the attribution and deletion target.
- Silence replacement preserves the waveform length.
- SpeechXAI’s WhisperX transcription is the only source of word text and timing; no character-based or uniform timing approximation is used.
- Transcription is performed once, and the returned word segments are reused by the explainer and plot so words, timestamps, and scores remain index-aligned.
- WhisperX’s per-word transcription confidence is retained in the attribution table so uncertain alignments remain visible during interpretation.
- The visualization distinguishes the lexical word interval from SpeechXAI’s effective mask, which extends 100 ms before and 40 ms after each word.
- Attribution bars use SpeechXAI’s raw confidence-drop values. They are not rescaled to make the largest word equal to one, because that would hide the magnitude of the model response.
- The waveform-loading path used by the project is checked against SpeechXAI’s pydub-loading path. Both must produce the same predicted class and probability vector before the explanation is accepted.
- Word intervals must be chronological, have positive duration, remain inside the audio, and correspond one-to-one with SpeechXAI’s returned features and scores.

- The notebook rejects an utterance when its first aligned word begins before SpeechXAI’s 100 ms pre-word padding. The upstream removal function does not clamp this slice at zero, so accepting such an example could make the displayed mask differ from the perturbation actually evaluated.
- The final evaluation uses the top- k SpeechXAI words to define the shared masking duration.
- Overlapping padded masks are counted by their union duration so the audio-removal budget is not inflated.
- Notebook 01 writes no result files; all results remain visible in the executed notebook.

What was tested

The selected audio was loaded and classified, then transcribed and aligned with SpeechXAI’s WhisperX function. The returned segments were passed directly into SpeechXAI’s leave-one-out explainer and checked against the returned feature order. Timestamp validity, word-score alignment, and the 100 ms first-word safety condition were validated. The original prediction obtained from SpeechXAI’s pydub representation matched the prediction obtained from the waveform shown in the notebook. The visualization used real word locations, effective silence masks, raw confidence-drop attribution values, and per-word transcription confidence. Top- k SpeechXAI deletion was compared with repeated random masking using the same union duration.

Outcome

The notebook executed successfully for utterance `Ses01F_impro01_F005`. WhisperX located the first word at 0.594 seconds, preserving the initial silence and later pauses. The returned word-confidence values ranged from 0.760 to 0.980. The classifier predicted anger with confidence 0.8781. SpeechXAI assigned the largest raw confidence drop, 0.4218, to “what’s”. The visualization, attribution table, consistency checks, and shared-duration evaluation all completed successfully.

Remaining issues

The current validation covers one utterance only. WhisperX confidence is useful diagnostic information but is not a guarantee that every boundary is correct. Utterances with a word beginning inside the first 100 ms are currently rejected because of the upstream SpeechXAI slicing behavior; supporting them safely would require clamping the library’s perturbation boundary. The pipeline must be run on a controlled multi-audio set before drawing comparative conclusions.

3.2 Notebook 02 — Attention Access Test

Purpose

Determine whether the wav2vec 2.0 Transformer attention maps required by the LeGrad-inspired method are accessible.

Technical idea

The active classifier is a Hugging Face `Wav2Vec2ForSequenceClassification` model. The diagnostic step requests attention outputs directly from its Transformer encoder and inspects their dimensions and numerical behavior.

Decisions made

- Attention access is treated as a required checkpoint before developing LeGrad.
- The Hugging Face classifier is accessed directly, matching the model-loading path used by SpeechXAI.
- Attention tensors should be inspected across layers and heads and visualized for at least one layer-head pair.

What was tested

The configured SUPERB classifier was run on the same IEMOCAP utterance used in Notebook 01 with attention output enabled. The notebook inspected every returned layer, validated finite values, printed tensor dimensions, and visualized one final-layer attention head.

Outcome

Attention access succeeded. The model returned 12 attention tensors, one per Transformer layer, each shaped [1, 12, 201, 201] for the selected utterance.

Remaining issues

Notebook 03 must confirm that these returned attention tensors retain usable, nonzero gradients for the original predicted-class logit.

3.3 Notebook 03 — Attention Gradient Test

Purpose

Verify that gradients of a target class score with respect to Transformer attention maps can be obtained.

Technical idea

LeGrad uses sensitivity to attention maps as its explanation signal. For speech, the target quantity is the classifier score for the original predicted emotion, and the required diagnostic is $\partial s_{\hat{y}} / \partial A_l$ for one or more Transformer layers.

Decisions made

- The target is the original predicted-class logit or score.
- Returned attention tensors will retain gradients during backpropagation.
- Gradient shape, magnitude, finite values, and nonzero proportion will be checked before attribution aggregation.
- Logits are preferred over probabilities when probability saturation weakens gradients.

What was tested

The notebook scaffold and reusable gradient-retention utilities are present.

Outcome

The conceptual gradient path is defined. A real model forward and backward pass remains to be validated after Notebook 02 succeeds.

Remaining issues

Gradient availability depends on how the wav2vec 2.0 attention implementation returns intermediate tensors. Hooks may be required if returned attentions are detached from the computational graph.

3.4 Notebook 04 — LeGrad-Inspired Temporal Explanation

Purpose

Convert attention-gradient information into a temporal relevance signal over the speech utterance.

Technical idea

For each Transformer layer, positive class-sensitive attention gradients are combined with attention activations, averaged across heads, aggregated across layers, and reduced from a token-token matrix to token relevance. Token relevance is then mapped to audio time.

Decisions made

- The method is presented as LeGrad-inspired because it adapts the central attention-gradient principle from vision to speech.
- Relevance uses positive gradients multiplied by attention values.
- Head and layer aggregation use simple means for interpretability and reproducibility.
- Temporal outputs use small fixed-duration bins or token intervals that can participate in duration-matched deletion evaluation.

What was tested

Reusable aggregation and token-to-time utilities have been prepared.

Outcome

The representation and output format are defined. Real relevance curves depend on successful attention-gradient extraction in Notebook 03.

Remaining issues

The reduction direction of the token-token relevance matrix must be selected empirically and kept fixed. Temporal mapping must also account for the wav2vec 2.0 feature encoder stride.

3.5 Notebook 05 — SpeechXAI Explanations

Purpose

Generate word-level explanation intervals for the same classifier inputs used by the LeGrad-inspired method.

Technical idea

SpeechXAI aligns speech with transcript words and measures the effect of perturbing each word-level audio segment. The quantitative comparison will use word identity, timestamps, and importance scores.

Decisions made

- Word-level explanations are the primary SpeechXAI output for quantitative comparison.
- Paralinguistic explanations remain available for qualitative analysis after the main word-level pipeline is stable.
- SpeechXAI output will use the common columns `audio_id`, `word`, `start`, `end`, and `score`.
- The same audio and original predicted class will be used across explanation methods.

What was tested

Notebook 01 ran SpeechXAI's WhisperX alignment and leave-one-out explainer on one IEMOCAP utterance using the same SUPERB classifier selected for the project.

Outcome

The single-audio SpeechXAI integration is operational. Real word timestamps, effective mask intervals, and attribution scores are available in the shared interval representation.

Remaining issues

Notebook 05 should extend the validated single-audio workflow to the controlled evaluation set and define how transcription and explanations are cached for repeatable dataset-scale runs.

3.6 Notebook 06 — Unified Interval Representation and Masking

Purpose

Provide a shared representation and masking protocol for word intervals, LeGrad time bins, and random baselines.

Technical idea

SpeechXAI top- k words define a duration X . LeGrad and random selections are then constrained to the same total duration. All selected regions are replaced by silence without changing sample count.

Decisions made

- SpeechXAI is the source of the shared duration because its native explanation unit is a word.
- LeGrad regions are ranked time bins selected until duration X is reached.
- Random selections use the same bin resolution and duration X .
- The last selected bin may be clipped to match X closely.
- Primary conditions are SpeechXAI-top, LeGrad-top, and Random.

What was tested

Duration-based interval selection, top- k selection, random interval sampling, silence masking, and union-duration accounting were tested. Notebook 01 matched the effective SpeechXAI and random mask durations for every evaluated k .

Outcome

The corrected duration-matching logic is available and the notebook scaffold executes successfully.

Remaining issues

The full comparison still requires LeGrad explanation intervals for the same utterances.

3.7 Notebook 07 — Deletion Evaluation

Purpose

Measure explanation faithfulness under the shared-duration masking protocol.

Technical idea

For every audio and top- k setting, the classifier confidence for the original predicted class is measured before and after masking. A faithful selection should remove regions that cause a larger confidence drop than same-duration random regions.

Decisions made

- Top- k values are 1, 2, 3, and 5.
- The main metric is absolute confidence drop for the original predicted class.
- Relative confidence drop and prediction flip rate are secondary metrics.
- Random masking is repeated 20 times to estimate its mean and variability.
- Each result records k , method, actual masked duration, original confidence, masked confidence, confidence drop, relative drop, flip state, and random-trial index.

What was tested

Notebook 01 validated the evaluation logic on one utterance using real SpeechXAI leave-one-out importance and repeated random masking. Notebook 07 currently executes as a protocol scaffold.

Outcome

The evaluation schema and fairness constraints are defined and validated at baseline scale.

Remaining issues

The complete comparison requires SpeechXAI and LeGrad explanations across a controlled multi-audio evaluation set.

3.8 Notebook 08 — Qualitative Examples

Purpose

Create interpretable examples that show where SpeechXAI and LeGrad agree or disagree and how masking changes model confidence.

Technical idea

Each example combines the waveform, word-level SpeechXAI intervals, the LeGrad temporal relevance curve, selected masking regions, and before/after classifier confidence.

Decisions made

- Examples will be selected from quantitative outcomes rather than chosen only for visual appeal.
- The final set should include agreement, LeGrad-favored, SpeechXAI-favored, and difficult cases.
- Interpretations will distinguish model faithfulness from human plausibility.

What was tested

The notebook structure and plotting requirements have been defined.

Outcome

The qualitative analysis plan is ready.

Remaining issues

Representative cases can be selected after Notebook 07 produces full deletion results.

3.9 Notebook 09 — Final Results Summary

Purpose

Aggregate deletion outcomes into the tables and plots used to answer the main research question.

Technical idea

Results are grouped by method and SpeechXAI top- k value. The analysis reports confidence-drop statistics together with mean masked duration, ensuring that interpretation includes the actual amount of removed speech.

Decisions made

- The horizontal axis of the deletion curve is SpeechXAI top- k , rather than an arbitrary audio-percentage budget.
- Random trials are summarized by their mean and standard deviation.
- Mean masked duration is reported for every k .
- Optional class-wise analysis will be added if the final sample size supports it.

What was tested

The notebook executes and handles the current absence of full deletion results gracefully.

Outcome

The aggregation schema and plot definitions reflect the corrected evaluation design.

Remaining issues

Final tables and plots require completed SpeechXAI and LeGrad runs.

4 Evaluation Design

The final quantitative evaluation follows these steps:

1. SpeechXAI returns importance scores for word-aligned audio segments.
2. For each audio and each $k \in \{1, 2, 3, 5\}$, the top- k SpeechXAI words are selected.
3. The selected words define a total duration X seconds.
4. Those SpeechXAI word intervals are replaced by silence, and the confidence drop for the original predicted class is measured.
5. LeGrad time bins are ranked by relevance, and the highest-ranked bins whose total duration equals X are selected.
6. Those LeGrad regions are replaced by silence, and the same confidence drop is measured.
7. Random time bins with total duration X are selected repeatedly to form a stable random baseline.
8. All methods use the same audio, classifier, target class, masking operation, and removed duration.

SpeechXAI naturally explains words, while the LeGrad-inspired method explains Transformer time bins or tokens. Equalizing the total occluded duration makes the comparison focus on where each method assigns relevance.

The main metric is:

$$\text{drop} = p_{\text{original}}(\hat{y}) - p_{\text{masked}}(\hat{y}),$$

where $\hat{y}$ is the class predicted from the original audio. Secondary metrics are relative confidence drop and prediction flip rate. Mean masked duration is reported for every top- k setting.

5 Methodological Decisions Summary

Component	Decision	Reason
Dataset	IEMOCAP speech, transcripts, and emotion labels	It is a standard speech emotion resource and provides the audio and annotations required by the selected classifier.
Classifier	SUPERB wav2vec2-base-superb-er checkpoint used by SpeechXAI	Using the same classifier removes model choice as a confound when comparing SpeechXAI and LeGrad-inspired explanations.
Model label space	Anger, happiness/excitement, neutral, sadness	These are the four outputs supported by the selected pretrained classifier.
Target class	Original predicted class	Faithfulness is evaluated relative to the model decision being explained.
SpeechXAI unit	Word-aligned audio segments	This is SpeechXAI’s native, human-readable explanation unit.
LeGrad unit	Transformer tokens or fixed temporal bins	Attention-gradient relevance is naturally produced at the encoder-token level.
LeGrad formula- tion	Positive attention gradients combined with attention values, aggregated across heads and layers	This preserves the central LeGrad idea while adapting its output to speech time.
Shared duration	X seconds from SpeechXAI top- k words	This makes word-level and time-bin explanations comparable under equal audio removal.
Top- k values	1, 2, 3, and 5 words	These settings provide a compact deletion curve while respecting SpeechXAI’s native output.
Masking method	Silence replacement with unchanged waveform length	It provides a simple perturbation and avoids duration changes that could independently affect the classifier.
Random baseline	Same-duration random bins, repeated 20 times	Repetition estimates baseline variance and provides a stable reference.
Main metric	Confidence drop for the original predicted class	It measures how strongly the selected regions support the model’s original decision.
Secondary metrics	Relative confidence drop and prediction flip rate	They complement absolute confidence drop across samples with different confidence levels.
Notebook 01 scope	One utterance, in-memory outputs	It establishes an understandable baseline and avoids mixing pipeline validation with dataset-scale evaluation.